Group 4 Network and Protocol Comments

Three network-related issues or extensions are considered below. Each comment identifies the observed behavior, the required processing change, and the relevant IRC protocol context.

1. Output backpressure and multi-line replies

The submitted server implements this network extension in `server/client_session.py`, `server/server.py` and `server/handlers.py`. A short JOIN can produce many JOIN, NAMES and terminator messages faster than a session writer drains them, so the application queue needs backpressure rather than treating a temporary queue burst as peer failure. The implementation waits for output headroom outside the shared IRC state lock, preserves message order, and uses a bounded timeout for a genuinely stalled reader; PONG and QUIT remain responsive. This changes JOIN/NAMES processing and numerics 353/366. Each IRC line is still limited to 512 bytes, while TCP delivery may contain many messages.

2. Presence queries and complete bounded replies

The submitted server and Bot implement this complete-result policy in `server/handlers.py` and `bot/queries.py`. An ISON request may fit the 512-byte client-line limit while its complete 303 reply becomes too long after the prefix, recipient nickname and CRLF are added. The server measures the encoded complete reply and returns one 417 ISON error instead of silently omitting online names. The Bot batches by both request and reply budgets, drains failed batches, and discards partial results on timeout or reconnect. The corresponding IRC commands are ISON and numeric 303; 417 is the project's explicit oversized-result error.

3. Registration state, nickname reservation and liveness

The submitted server implements this registration/liveness policy in `server/server.py`, `server/state.py` and `server/handlers.py`. A client can send NICK, reserve a nickname and answer PING/PONG without completing USER, so the server timestamps connection creation and enforces a monotonic deadline that activity cannot extend. Expiry removes the client and nickname atomically; private routing and presence visibility require completed registration. This follows the IRC NICK/USER registration sequence and PING/PONG liveness messages: a responsive TCP peer is not necessarily a registered IRC client.

References: RFC 2812, sections 3.2.2 (JOIN), 3.2.5 (PART), 3.2.6 (LIST), 3.4.2 (LUSERS), 3.7.2-3.7.3 (PING/PONG), 4.9 (ISON), and 5.1 (numeric replies).